

DEVARSHEE SANDILYA

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Profile Summary

Senior Data Scientist with 4+ years of experience building and deploying production-grade Machine Learning and Generative AI systems. Specialized in LLM and Agentic AI applications, Retrieval-Augmented Generation (RAG), and Snowflake-based data engineering. Proven track record of delivering measurable business impact, including 20x increase in sales opportunities. Strong expertise in Python, scalable data pipelines, and end-to-end AI system deployment.

Technical Skills

Programming: Python, SQL

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost, Random Forest, Supervised Learning, Time Series

Generative AI and NLP: LangGraph, LangChain, LlamaIndex, Hugging Face, Prompt Engineering, Vector Databases, Azure OpenAI, Fine-tuning (PEFT)

Data Engineering: Snowflake, ETL Pipelines, UDF/UDTF, Streams, Tasks, Data Warehousing

Visualization Tools: Power BI, Streamlit, Flask, Tableau

Cloud Platforms: Azure, Google Cloud Platform

Other Tools: OpenCV, OR-Tools, CPLEX, Git

Professional Experience

Donaldson Company, Inc

May 2023 - Present

Senior Data Scientist

Gurugram

- Designed and deployed fully automated, Snowflake-native data pipelines using UDF/UDTFs, Streams, and Tasks to process large-scale unstructured data, automating **40 hours per week** of manual effort
- Developed **LLM-powered** document intelligence systems using Snowflake Document AI and Cortex functions to extract structured sales signals (company names, geolocations, market segments)
- Built end-to-end Retrieval-Augmented Generation (**RAG**) pipeline using Snowflake Cortex functions and **LangChain**, enabling semantic search and enterprise knowledge retrieval
- Engineered scalable **ETL** pipelines integrating unstructured news data with structured datasets (XML/FTP), delivering consolidated insights through **Power BI** dashboards
- Researched and implemented advanced computer vision models (**Faster R-CNN**, **SAHI**, **YOLT**) for small-object detection, improving detection accuracy and reducing manual inspection efforts
- Deployed real-time **OpenCV**-based monitoring systems for industrial processes, reducing analysis time by **4 hours** per cycle

Analyttica Datalab

Jan 2022 - May 2023

Junior Data Scientist

New Delhi

- Built **ARIMA-based** time series forecasting models for EV battery performance, improving predictive maintenance capabilities
- Applied optimization techniques (linear programming, constraint optimization) using **OR-Tools**, **PuLP** and **CPLEX** for transportation and resource allocation problems
- Contributed to end-to-end ML workflows including data preprocessing, feature engineering, model training, and evaluation

Key Achievements

Sales Signal Intelligence Platform

(Snowflake Cortex Function, Stored Procedures, UDFs, LLM, NER)

- Automated Gen-AI powered end-to-end platform using Snowflake LLMs, RAG, LangChain to extract insights from online web articles

- Impact: Led to high-volume news data, resulting in a **20x** increase in identified sales opportunities

Computer Vision Systems

(OpenCV, Python, Azure Devops, Azure Web App)

- Created real-time OpenCV-based application for tracking and analyzing particle motion in fluids
- Implemented advanced image processing techniques to improve defect identification accuracy by **5 percent** , deployed as a scalable Azure WebApp service

Certifications

Coursera: Agentic AI with LangChain and LangGraph

(Coursera, LLama 3.3, ChromDB, DocLing)

- Created a ReAct based agent using LangGraph for recommending tourist places, activities and dress for diverse people using **Tavily Expert API** as web search tool
- Developed a multi-agent RAG system **DocChat** for querying medical documents with Hybrid Retriever(**BM25**) and self-correcting mechanism and deployed as **Gradio** app

Core Competencies

Machine Learning, Generative AI, Agentic AI, NLP, Data Engineering, Computer Vision, Model Deployment, Model Validation, Feature Engineering, Stakeholder Management, Cross-Collaboration, ETL Pipelines

Education

University of Edinburgh

Master of Science in Theoretical Physics

September, 2018 – August, 2019

Edinburgh, UK

University of Pune

Bachelor of Science in Physics

July, 2015 – May, 2018

Pune, India